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EXPERIMENTAL STUDY AND CONJUGATE CFD OF HEAT TRANSFER IN AN ANNULAR MINICHANNEL

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This work presents an experimental investigation supported by conjugate CFD simulations of refrigerant flow in a 1 mm-deep annular minichannel. The test section was electrically heated using an internally mounted, axially aligned heater, and the imposed heat flux was determined from the measured electrical power. Bulk fluid temperature and pressure were monitored at the inlet and outlet together with the mass flow rate, while the heater wall temperature was recorded at selected locations using thermocouples under steady-state conditions. A conjugate heat transfer model was developed in Simcenter STAR-CCM+ to reproduce the geometry and operating conditions of the experiment. The numerical setup accounted for heat conduction in the solid and convection in the fluid, with refined near-wall resolution in the annular gap. Model performance was assessed by comparing predicted wall temperatures at thermocouple locations with the corresponding measurements. The study also examined how modelling choices influence the results, providing guidance for reliable CFD predictions in annular minichannels.